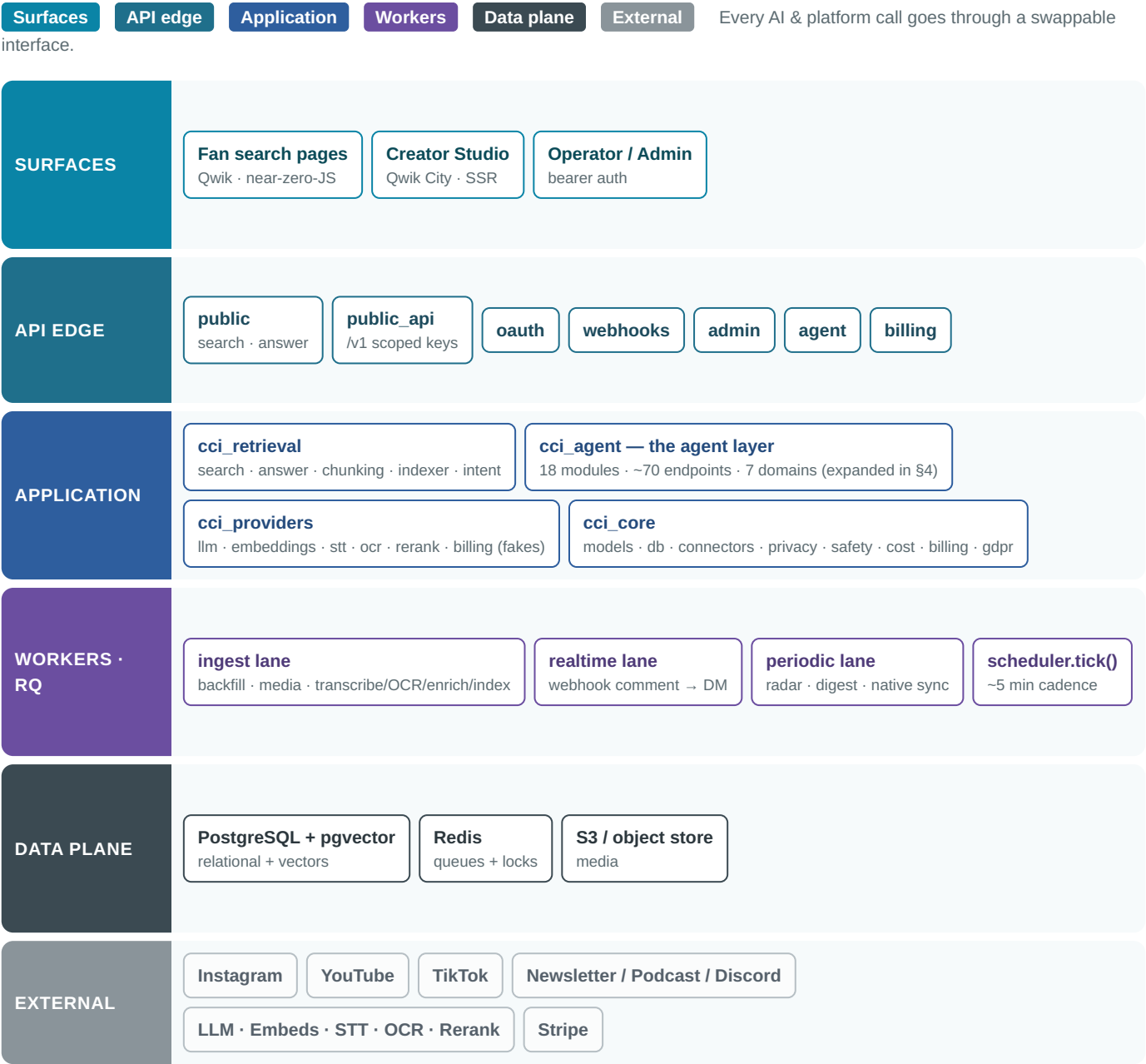


Sift — Architecture & Workflow Pipeline

A cross-platform demand-to-outcome platform for creators · system overview and end-to-end data flow

1 · System architecture (layered)



2 · The product loop

The company is one cycle — v1 *listens & serves*; the agent layer *advises, executes, closes*. The loop closes when the original asker hears back, seeding the next question.



3 · The complete workflow pipeline (end to end)

From a creator connecting an account to a closed, measured demand loop. Tags show the lane / surface each stage runs on.

- 1 Connect** oauth · admin
Creator OAuth or operator token; capability-tagged account (incl. TikTok per-account full-loop).
`oauth.py · admin.set_token · PlatformAccount · connectors registry`
- 2 Sync / Ingest** ingest lane
Manual 'Sync now', 'Sync all', or hourly. Connector output → Posts + Comments, idempotent, per-item SAVEPOINT, PII-redacted + pseudonymized.
`connectors.backfill_content → sync_native / ingest_instagram / ingest_youtube / ingest_import`
- 3 Enrich** ingest lane
Per new post: transcribe (Whisper), OCR on-screen text, LLM summary/topics/keywords, language detection.
`enrich.py · providers (stt, ocr, llm) · language.py → Transcript`
- 4 Index** ingest lane
Chunk content + embed into pgvector beside a generated tsv full-text column. Deletion purges chunks.
`chunking.py · indexer.py · embeddings provider → Chunk(embedding, tsv)`
- 5 Serve** API · public
Hybrid search: FTS + pgvector cosine → RRF fusion → optional rerank → confidence. Grounded cited answer card, or 'no strong answer'.
`search.py (RRF + rerank) · answer.py (cited; voice = post-retrieval rewrite)`
- 6 Listen** realtime lane
Comment webhook (signature-verified, idempotent) → intent detection → same retrieval layer. Query logged as demand.
`webhooks.py · intent.py · _claim_event (dedup)`
- 7 Respond** realtime lane
Approval-mode DmJob → dispatcher: capability gate + per-creator advisory lock + 7-day window + hourly cap. One public reply + one private DM (answer + deep link); low-confidence → archive link.
`dm.dispatch_approved · _can_dm · _try_lock · deep_links.py`
- 8 Learn** periodic lane
Cluster searches + comments + external signals → Demand Integrity (unique askers, organic/prompted, persistence, sentiment, manipulation risk, exposure-normalized) → explainable Opportunity Score.
`radar.build_radar · demand.compute_integrity · opportunity_score → DemandTopic`
- 9 Reconcile** API · admin
Each card classified against the creator's own archive: gap / partial / well, with the covering permalink — 'make new' vs 're-promote'.
`_reconcile_demand · coverage strength + permalinks`
- 10 Act** agent layer
One-tap: content brief + draft (script/hooks), series builder, multi-format repurpose + Sift & Shift, reply assistant, Loop-Closer, briefing.
`drafting · repurposing · engagement · briefing · canonical`
- 11 Close** state machine
Publish or re-promote → demand state → loop_closed; Outcome logged. North-star: closed loops/week, split made vs re-promoted.
`agent_foundations.transition_demand · repromote_topic · Outcome`
- 12 Measure** eval · cost
Citation-correctness gate (fails closed if nothing answerable) before automation; events + per-creator AI cost metering; causal layer (holdouts, baselines, CIs) measures lift.
`cci_eval.harness · events.py · cost.meter_llm · causal.py`

4 · Agent layer — feature architecture

`cci_agent` is the "chief of staff" — **18 modules, ~70 endpoints across seven domains**, all implemented and exposed via the `/agent/` router. Every chip below is a working feature.

Agent domains * = scaffolded (opt-in gate + privacy guarantee + response shape; placeholder aggregates until a multi-creator panel exists).

FOUNDATIONS §02	Demand clustering	Voice memory	Offer-awareness	Rules & preferences	Permission ladder	
	State machine	Outcome event log				
AUDIENCE & DEMAND §03	The Briefing	Content gap map	Persona segmentation	Sentiment & emotion	Trend detection	
	Cross-platform synthesis	Competitor radar*	Opportunity Score	Demand Certificate		
CONTENT PRODUCTION §04	Content briefs	Draft generator	Creator recall	Repurposing	Sift & Shift	Series builder
	Performance prediction	Content calendar	Thumbnails	Canonical answers	Claim layer	
ENGAGEMENT & INBOX §05	Intent-labelled queue	Bounded clarifying	No-answer waitlist	Saved playbooks	Reply assistant	
	Agentic DM	Community delegation	'Not now' queue	Loop-Closer		
MONETIZATION §06	Auto-affiliate injection	Offer-aware CTAs	Affiliate optimisation	Sponsor matchmaker + pitch		
	Dynamic pricing	Revenue forecasting				
BRAND & REPUTATION §07	Crisis / sentiment-shift	Voice consistency scoring	Plagiarism monitoring*			
GROWTH & STRATEGY §08	Content strategy advisor	Peer benchmarking*	Education & skill plan			
OPERATIONS §09	Task / PM export	Customer CRM	Team & roles	Public API + keys		
CAUSAL (MEASUREMENT)	Interventions	Holdouts	Matched baseline	Holdout lift	Confidence intervals	

Foundations §02 — the data & memory the agent rides on

Feature	Module · function	Endpoint
Demand clustering & dedup	radar.build_radar	admin .../radar/build
Voice memory	agent_foundations.capture_voice	POST/GET /creators/{id}/voice
Offer-awareness	monetization.offer_cta + offers	POST/GET /creators/{id}/offers
Rules & preferences	agent_foundations.get_rules	GET/PUT /creators/{id}/rules
Permission ladder	permissions.permission_for/ can_auto_execute	GET/PUT /permissions · / automation/pause
Demand-item state machine	agent_foundations.transition_demand	POST /demand/{id}/transition
Intent / outcome event log	Outcome · events	GET /creators/{id}/outcomes

§03 Audience & demand intelligence

Feature	Module · function	Endpoint
The Briefing	briefing.build_briefing	GET /creators/{id}/briefing
Content gap map	briefing.content_gap_map	GET /creators/{id}/gap-map
Persona segmentation	intelligence.segment_personas	GET /creators/{id}/personas
Sentiment & emotion	intelligence.score_sentiment	GET /creators/{id}/sentiment
Trend detection	intelligence.detect_trends	GET /creators/{id}/trends
Cross-platform demand synthesis	radar + demand sources	POST /creators/{id}/external-signal
Competitor demand radar (opt-in)	growth.competitor_radar	GET /creators/{id}/competitor-radar
Opportunity Score (explainable)	demand.opportunity_score	GET /demand/{id}/opportunity
Demand Certificate	demand.demand_certificate	GET /demand/{id}/certificate
Pipeline + north-star	demand.closed_loops	GET /demand/pipeline · /north-star

§04 Content production

Feature	Module · function	Endpoint
Content briefs	drafting.generate_brief	(via draft)
Draft generator (script + hooks)	drafting.generate_draft	POST /demand/{id}/draft · /drafts
Creator recall search	drafting.creator_recall	GET /creators/{id}/recall
Multi-format repurposing	repurposing.repurpose	POST /posts/{id}/repurpose
Sift & Shift	repurposing.sift_and_shift / draft_shift	GET /sift-and-shift · POST /posts/{id}/shift
Series builder	repurposing.build_series	POST /demand/{id}/series
Performance prediction	scale.predict_performance	GET /drafts/{id}/predict
Content calendar / planner	scale.content_calendar	GET /creators/{id}/calendar
Thumbnail & visual intelligence	scale.thumbnail_concepts	(via scale)
Canonical answers (cross-platform)	canonical.group_variants	POST/GET /creators/{id}/canonical
Versioned claim layer	claims.extract_claims / detect_contradictions	GET /claims · /claims/{id}/approve

§05 Engagement & inbox

Feature	Module · function	Endpoint
Intent-labelled queue	inbox.labelled_inbox	GET /creators/{id}/inbox
Bounded clarifying question	inbox.bounded_clarify	(retrieval-side)
No-answer waitlist	inbox.add_to_waitlist	(fan-side)
Saved playbooks	inbox.match_playbook	POST/GET /creators/{id}/playbooks
Reply assistant	engagement.draft_reply	(DM job draft)
Agentic DM (approval mode)	engagement.handle_dm_followup / bulk_approve	POST /creators/{id}/dm/bulk-approve
Community delegation	engagement.delegation_candidate	(DM dispatch)
'Not now, but...' queue	engagement.defer_question / due_deferrals	POST /defer · GET /deferrals/due
Loop-Closer	engagement.close_loop	POST /demand/{id}/close-loop

§06 Monetization & revenue

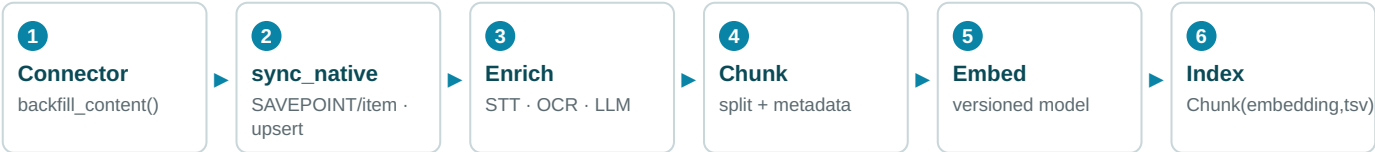
Feature	Module · function	Endpoint
Auto-affiliate injection	monetization.inject_affiliate / with_utm	(caption-time)
Offer-aware CTAs	monetization.offer_cta	(answers/recs)
Affiliate optimisation	revenue.affiliate_optimisation	GET /creators/{id}/affiliate-optimisation
Sponsor matchmaker & pitch	revenue.sponsor_report	GET /creators/{id}/sponsor-report
Dynamic pricing & offer insights	scale.pricing_insights	GET /creators/{id}/pricing-insights
Revenue forecasting & goals	scale.revenue_forecast	GET /creators/{id}/revenue-forecast

§07 Brand & reputation · §08 Growth & strategy · §09 Operations

Feature	Module · function	Endpoint
Crisis / sentiment-shift detection	brand.detect_crisis	GET /creators/{id}/crisis-check
Voice consistency scoring	brand.voice_consistency_score	POST /creators/{id}/voice-score
Plagiarism monitoring (opt-in)	growth.plagiarism_scan	POST /creators/{id}/plagiarism-scan
Content strategy advisor	intelligence.strategy_advisor	POST /creators/{id}/strategy
Peer benchmarking (opt-in)	growth.peer_benchmark	GET /creators/{id}/benchmark
Education & skill planning	scale.education_plan	GET /creators/{id}/education-plan
Task / project export	growth.export_task	POST /demand/{id}/export-task
Customer CRM	crm.customer_profiles / lifecycle_segments	GET /creators/{id}/customers · /lifecycle
Team & role-based access	team.add_member / has_capability / audit	POST/GET /creators/{id}/team
Public API & dev ecosystem	team.mint_api_key + public_api	POST /api-keys · GET /v1/public/*
Causal: interventions / lift / CIs	causal.create_intervention / holdout_lift	GET /interventions · /lift

5 · Key sub-pipelines

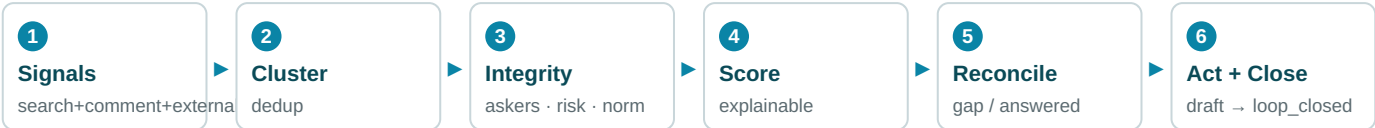
A · Ingestion — content becomes searchable



B · Retrieval & answer — one hybrid layer serves search, the answer card, and DM intent



C · Demand → action loop — the monetizable asset



6 · Jobs, queues & scheduler

Lane	What runs	Cadence
ingest	Backfills, media download, transcribe / OCR / enrich / index	On connect & on demand
realtime	Webhook comment → intent → DM job; approved-DM dispatch (advisory-locked)	Webhook + every tick
periodic	Demand Radar build, weekly digest, incremental + native sync, token refresh	Scheduled

scheduler.tick() (~5 min): DM dispatch every tick · hourly incremental + native sync · weekly radar + digest (Mon 07:00 UTC) · daily OAuth token refresh (06:00 UTC).

7 · Cross-cutting concerns

Concern	How it's enforced
Security	Creator-scoped queries; OAuth tokens encrypted at rest (EncryptedString); fail-closed boot on insecure secrets; webhook signature verified (fail-closed in prod); prompt-injection hardening (wrap_untrusted + GUARD) on agent prompts; DM dispatch advisory lock.
Privacy / GDPR	Per-creator pseudonyms (dedicated secret); PII redaction before storage; export + erase (incl. erase-by-pseudonym); opt-in row-level security.
Cost	Per-creator AI metering with op tags; per-plan monthly cost cap; separate anonymous-answer abuse ceiling (free card stays free).
Portability	Every AI + platform call behind an interface (offline fakes by default); Postgres + pgvector; S3; Alembic migrations 0001–0004; k8s + Render manifests.
Quality	Citation-correctness eval gate before automation; 311 backend + 9 frontend tests; five-dimension adversarial review.